
How Does Geometric Bias Shape the Denoising Trajectory in Flow-Based Protein Structure Generation?

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Abstract

Proteina is a flow-based generative model for protein backbone structure where geometry enters only as an additive pair bias B to attention scores. We study when and how this bias becomes essential during generation, using four complementary lenses across four model scales (60M–400M parameters). Our central finding is a sharp temporal asymmetry: providing B only during early generation ($t \leq 0.5$) then removing it is indistinguishable from never having B at all, while providing B only during late generation preserves structural quality. This pattern replicates identically across all four models and is invariant to protein length, despite order-of-magnitude variation in the descriptive geometry–content balance. Replacing B with magnitude-matched random noise is equally catastrophic as zeroing it, confirming the model depends on the specific geometric *information* in B , not merely its presence. Triangle multiplicative updates do not change *when* B is essential but invert *where*: they shift B ’s critical contribution from early to deep layers.

1 Introduction

Generative models for protein backbone structure have advanced rapidly, with flow matching emerging as a leading paradigm [8]. Proteina [5] scales flow-based protein generation to state-of-the-art quality by combining a transformer backbone with *Pair Bias Attention* — a mechanism that fuses learned content scores with geometric pair biases.

Unlike SE(3)-equivariant networks [4, 11] or invariant point attention [6], Proteina’s transformer layers are standard (non-equivariant) attention and MLP blocks. Geometry enters only through an additive pair bias B derived from pairwise distances and sequence separation — a *soft invariant prior* that encodes geometric information without constraining the network’s outputs to be equivariant. This reflects a broader trend away from hard-coded architectural symmetries toward soft geometric priors, as seen in ViT [3], TokenGT [7], and AlphaFold3 [1].

This raises a natural question: **when does the geometric bias become essential during generation, and what role does it play?** During generation, Proteina samples via SDE integration from pure noise ($t=0$) to a structured backbone ($t=1$). At each step, every attention layer computes:

$$A = \text{softmax}(C + B) \tag{1}$$

where $C = QK^\top / \sqrt{d}$ is the content score and B is the pair bias.

Importantly, B is constructed from binned pairwise distances of the *current* noisy structure $\mathbf{x}_t$. At $t=0$, $\mathbf{x}_t$ is random noise and these distances are uninformative; at $t=1$, they encode the near-final structure. This means that *static* comparisons of B and C at late timesteps are partly tautological — B encodes distances by construction, so its superiority at distance-related tasks near $t=1$ is expected. The more interesting questions concern the *dynamics*: when does B transition from uninformative noise to essential geometric signal, and how does this depend on model scale and architecture?

Since softmax is nonlinear, there is no single correct way to attribute the output to B vs. C . We therefore ask four complementary questions:

1. **Descriptive:** How large is B relative to C in the pre-softmax scores?
2. **Functional:** Which component produces more structurally useful attention?
3. **Causal:** When and where is B necessary for valid structures?
4. **Representational:** Where in the network does 3D structure emerge?

We apply all four lenses to four model variants spanning 60M to 400M parameters (Table 1). Our main findings are:

- **Sharp temporal asymmetry:** B is dispensable during early generation but essential during late refinement. This is not a gradual transition — removing B after *any* point produces collapse; providing B only during the last portion always preserves structure (Section 3.1).
- **Scale and length invariance:** The temporal pattern replicates across all four models and across protein lengths $n=50-200$, despite order-of-magnitude variation in the descriptive signal (Section 3.1).
- **Information, not magnitude:** Replacing B with magnitude-matched random noise is equally catastrophic as zeroing it (Section 3.1).
- **Architecture determines where, not when:** Triangle updates invert which layers are critical for B without changing the temporal pattern (Section 3.1).
- **Geometric interpreter layers:** Every model concentrates pre-softmax geometry dominance in one or a few layers, but the location varies across models (Section 3.3).

2 Methods

2.1 Models and Generation

We analyse four Proteina variants (Table 1). The no-tri models use only pair bias attention; the tri models additionally use triangle multiplicative updates that iteratively refine the pair representation. All models include 10 register tokens [2]. Proteins of length $n=100$ are generated unconditionally via 100-step SDE sampling (Euler-Maruyama with $dt=0.01$). Descriptive and functional experiments use 5 random seeds; causal ablations use 3 seeds. Error bars report ± 1 standard deviation.

Table 1: Model variants analysed.

Model	Version	Layers	Heads	d_{token}	d_{pair}	Triangle
60M no-tri	v1.3	12	12	512	256	No
200M no-tri	v1.2	15	12	768	256	No
200M tri	v1.1	15	12	768	512	Yes
400M tri	v1.4	18	16	1024	512	Yes

2.2 Pair Bias Construction

The pair bias B is constructed from three inputs concatenated into a 319-dimensional feature vector per residue pair: (i) binned pairwise $C\alpha$ distances from the current noisy structure $\mathbf{x}_t$ (64 bins, 0.1–3.0 nm), (ii) binned distances from self-conditioning predictions $\mathbf{x}_{sc}$ (128 bins), and (iii) binned sequence separation $|i-j|$ (127 bins). These features are linearly projected to d_{pair} dimensions (Table 1), layer-normalised, then projected to H scalar biases (one per attention head) via a learned linear layer.

In no-tri models (60M, 200M no-tri), B is computed once from the initial pair features and remains *constant across all layers*. In tri models (200M tri, 400M tri), the pair representation is updated every 3–5 layers via triangle multiplicative updates that incorporate information from the evolving sequence representation, so B changes through the network.

2.3 Metrics

Table 2 defines all metrics with null baselines. The primary structural quality metrics are:

- **Radius of gyration** (R_g): measures compactness. Well-folded proteins of length $n=100$ have $R_g \approx 1.2\text{--}1.5\text{ nm}$; collapsed structures have $R_g < 0.2\text{ nm}$ [5].
- **Steric clashes**: count of $C\alpha$ pairs with distance $< 2\text{ \AA}$ and $|i-j| \geq 3$, representing physically impossible atomic overlaps. Valid structures have zero clashes; thousands of clashes indicate structural collapse.

Table 2: Summary of metrics. “Baseline” is the expected value under a null model.

Metric	Definition	Range	Baseline
<i>Descriptive</i>			
$R = \ B\ _F / \ C\ _F$	Raw Frobenius norm ratio of the pre-softmax bias and content matrices.	$[0, \infty)$	$R = 1$ (balanced)
$R_c = \ \tilde{B}\ _F / \ \tilde{C}\ _F$	Row-centred variant: $\tilde{X}$ subtracts each row’s mean. Removes softmax-invariant components (since $\text{softmax}(\mathbf{x} + c) = \text{softmax}(\mathbf{x})$).	$[0, \infty)$	$R_c = 1$ (balanced)
<i>Functional</i>			
Precision@ $L/5$	Fraction of the top- k attention-ranked residue pairs ($k = n/5$) that are true contacts ($C\alpha < 8\text{ \AA}$, $ i-j \geq 6$). Symmetrised and APC-corrected following Rao et al. [10].	$[0, 1]$	Contact density ($\sim 1\text{--}3\%$)
<i>Causal</i>			
R_g	Radius of gyration (see above).	$(0, \infty)\text{ nm}$	$\sim 1.2\text{--}1.5\text{ nm}$
Steric clashes	$C\alpha$ pairs with $d < 2\text{ \AA}$, $ i-j \geq 3$.	$[0, \infty)$	0
<i>Representational (structure lens)</i>			
RMSD to final	Kabsch-aligned RMSD of decoded intermediate-layer coordinates to the final-layer output.	$[0, \infty)\text{ nm}$	0 at final layer
Contact similarity	Jaccard index of contact maps ($< 8\text{ \AA}$, $ i-j \geq 6$) between intermediate and final layer.	$[0, 1]$	1 at final layer

2.4 Analytical Framework

Lens 1: Descriptive. We measure R_c , the row-centred Frobenius norm ratio (Table 2). Row-centering removes energy that is invariant under softmax: since $\text{softmax}(\mathbf{x} + c) = \text{softmax}(\mathbf{x})$, row-constant components inflate the raw ratio R without affecting attention. Figure 5 shows this correction can change R by $6\times$. We also decompose R_c by sequence separation to test whether B ’s role differs for local vs. long-range pairs.

Lens 2: Functional. Following Rao et al. [10], we evaluate Contact Precision@ $L/5$ for $\text{softmax}(C + B)$, $\text{softmax}(B)$, and $\text{softmax}(C)$. Ground truth contacts are defined retrospectively from each model’s own generated structure — an important caveat discussed in Section 3.2. Short-range pairs ($|i-j| < 6$) are excluded because neighbouring residues along the backbone are always spatially close regardless of the protein’s fold. The random baseline is the contact density among eligible pairs (the precision expected from uninformative attention rankings).

Lens 3: Causal. We zero out B under six conditions: full ablation, early-only B ($t \leq 0.5$), late-only B ($t > 0.5$), Layer-0 ablation, deep-layer ablation (second half of layers), and baseline. We also ablate each model’s peak- R_c layer individually, and perform a fine-grained temporal sweep (split at $t = 0.2, 0.3, \dots, 0.8$). Additionally, we replace B with magnitude-matched Gaussian noise to distinguish whether the model requires B ’s geometric *information* or merely its signal *magnitude*.

Lens 4: Representational (structure lens). Inspired by the logit lens [9], we apply the coordinate decoder to intermediate layer representations, producing a per-layer 3D prediction compared to the final output via RMSD and contact similarity.

3 Results

3.1 Causal: B Is Essential Only During Late Generation

We present the causal results first because they are the least affected by the circularity concern noted in the introduction (that B encodes distances by construction).

Table 4 shows structural quality under ablation. Full removal of B is catastrophic in every model: structures collapse to $R_g < 0.15$ nm with thousands of steric clashes (vs. $R_g \approx 1.3$ nm and 0 clashes at baseline).

Sharp temporal asymmetry. Giving B only during early generation ($t \leq 0.5$) then removing it is indistinguishable from full ablation in all four models (Table 4, rows 3–4). A fine-grained temporal sweep (Figure 1) reveals this is not a “cliff” at a specific timestep but a one-sided asymmetry: removing B after *any* point (including $t=0.2$) produces collapse, while providing B only during the last portion (even just $t > 0.8$) always preserves valid structures, albeit with reduced compactness.

The model has no memory of early B . The temporal asymmetry could arise because (a) only late B matters, or (b) removing B after the model has seen it destabilises the trajectory. A “gap” experiment distinguishes these: B is active for $t \in [0, 0.3] \cup [0.7, 1]$ (on, then off, then on again). If the gap structures are comparable to late-only controls (B on only for $t > 0.7$), then early B is truly irrelevant; if worse, then removing B after showing it causes additional damage. Table 3 shows the gap conditions match the late-only controls within 0.02 nm in both models tested. The model does not retain any benefit — or harm — from B ’s early presence; only the current availability of B matters.

Table 3: Gap experiment ($n=100$, 3 seeds). Gap conditions (B on $\rightarrow$ off $\rightarrow$ on) match late-only controls.

Condition	60M no-tri		400M tri	
	R_g	Clash	R_g	Clash
Baseline	1.44	0	1.27	1
Gap $[0,0.3] \cup [0.7,1]$	1.10	0	1.05	1
Late-only $[0.7,1]$	1.11	0	1.04	1
Gap $[0,0.2] \cup [0.8,1]$	1.05	0	0.96	4
Late-only $[0.8,1]$	1.06	0	0.96	2

Scale and length invariance. The temporal pattern replicates identically across all four models (Table 4) and across protein lengths $n=50$, 100, and 200 (Table 7). The boundary does not shift with model scale or protein complexity, suggesting it is a property of the flow matching objective itself.

Geometric information, not magnitude. Replacing B with magnitude-matched Gaussian noise ($\mathcal{N}(0, \sigma_B^2)$, same standard deviation as the true B) is equally catastrophic as zeroing B across all four models (Table 5). This rules out the hypothesis that B serves as a scale regulariser and confirms the model depends on the specific geometric content.

Triangle updates invert the layer-specific profile. In no-tri models, Layer-0 ablation is the most damaging single-layer intervention ($R_g = 0.81$ nm for 60M). In tri models, Layer-0 ablation is nearly harmless ($R_g \geq 1.1$ nm) while deep-layer ablation is catastrophic ($R_g \leq 0.12$ nm, thousands of clashes). The temporal pattern is identical in both cases.

No single layer is indispensable. Ablating each model’s peak- R_c layer (Table 6) produces only moderate damage: R_g drops by 5–18% but structures remain valid (0–2 clashes). The geometric bias functions as a distributed system — each layer’s contribution is individually replaceable, but the aggregate is essential.

Table 4: Structural quality under bias ablation ($n=100$, 3 seeds). Temporal asymmetry replicates across all four models. Bold: triangle update effect on layer-specific profile.

Condition	60M no-tri		200M no-tri		200M tri		400M tri	
	R_g	Clash	R_g	Clash	R_g	Clash	R_g	Clash
Baseline	1.44	0	1.36	0	1.42	0	1.27	1
Full ablation	0.12	3532	0.14	2968	0.07	4702	0.11	4021
Early-only B	0.12	3532	0.14	2967	0.07	4702	0.11	4021
Late-only B	1.19	0	1.19	4	1.12	2	1.17	1
Layer-0 abl.	0.81	1	0.91	5	1.24	1	1.13	0
Deep abl.	0.60	23	0.63	32	0.05	4752	0.12	3560

Table 5: Random bias swap ($n=100$, 3 seeds). Random B collapses structures just like zero B .

Condition	60M no-tri		200M no-tri		200M tri		400M tri	
	R_g	Clash	R_g	Clash	R_g	Clash	R_g	Clash
Baseline	1.44	0	1.36	0	1.42	0	1.27	1
$B = 0$	0.12	3532	0.14	2968	0.07	4702	0.11	4021
$B = \mathcal{N}(0, \sigma_B^2)$	0.10	4181	0.18	1861	0.08	4648	0.12	3675

3.2 Functional: B Carries the Structural Signal

Figure 2 evaluates contact prediction across all four models. At $t=1$, B -only precision is 5–20 $\times$ above the random baseline: 0.26 for the 60M and 200M no-tri models, 0.39 for 200M tri, and 0.46 for 400M tri (random ≈ 0.01 –0.03). C -only precision is only 2–3 $\times$ above random (~ 0.05 –0.10).

Circularity caveat. Because B encodes pairwise distances and contacts are defined by pairwise distances, B -only’s superiority at late t is partly expected by construction. The more informative observation is that C — which has access to the full forward pass — never learns to encode structural information that B already provides. Because all models were trained *with* B present, C was never pressured to learn redundant geometric features. This is an instance of shortcut learning: B ’s availability during training shaped C ’s role, rather than C being inherently incapable of encoding structure. Nevertheless, the functional gap provides a useful validation that B carries the expected geometric content and that C has specialised for non-structural roles.

Full attention slightly underperforms B -only, suggesting C occasionally opposes B by encoding complementary non-contact relationships.

3.3 Descriptive: The Geometry–Content Balance

Figures 3 and 4 show R_c across the generation trajectory.

Every model concentrates geometry dominance in one or a few layers (Figure 4). The 60M model concentrates in Layer 0 ($R_c \approx 35$ at $t=1$); the 200M no-tri model in Layers 0 and 2 ($R_c \approx 4$); the 400M tri model in Layer 2 ($R_c \approx 29$). The 200M tri model is distinctive: it distributes high R_c across many layers (L4, L7, L9, L13), with L9 peaking at $R_c \approx 27$. The remaining layers in all models stay near $R_c \approx 1$.

The peak layer location varies across models. Both no-tri models peak in early layers (L0); the 400M tri model peaks at L2 (still early); only the 200M tri model peaks deep (L9). There is no clean trend with either scale or the presence of triangle updates. This contrasts with the causal layer-sensitivity profile, which *does* cleanly separate by architecture (Section 3.1): descriptive and causal measures tell different stories about which layers matter.

Why R_c rather than R ? The raw ratio $R = \|B\|_F / \|C\|_F$ does not account for softmax invariance. Figure 5 compares R and R_c for the 60M model: at $t=1$, $R \approx 5.7$ but $R_c \approx 35$. The content score C develops large row-constant components that inflate its Frobenius norm without affecting the attention distribution. R_c removes these by subtracting each row’s mean before computing the norm.

Table 6: Peak-layer ablation (3 seeds). No single geometric interpreter layer is catastrophically important.

Model	Peak layer	Peak R_c ($t=1$)	R_g (ablated)	R_g (baseline)	Clashes
60M no-tri	L0	34.9	0.81	1.44	1
200M no-tri	L2	3.9	1.28	1.36	1
200M tri	L9	27.0	1.17	1.42	1
400M tri	L2	29.4	1.13	1.27	1

Table 7: Temporal ablation across protein lengths (60M no-tri, 3 seeds). The pattern is length-invariant.

Condition	$n=50$		$n=100$		$n=200$	
	R_g	Clash	R_g	Clash	R_g	Clash
Baseline	1.15	0	1.44	0	1.96	0
Full ablation	0.16	487	0.12	3532	0.09	18116
Early-only B	0.16	487	0.12	3532	0.09	18117
Late-only B	0.97	0	1.19	0	1.52	0

Sequence separation modulates the balance. Figure 6 decomposes R_c by residue separation. The 60M model shows the strongest separation dependence; larger models are progressively more uniform. The 200M tri model again stands out with a distinctive non-monotonic trajectory.

3.4 Representational: Structure Emerges at the Output

Figure 7 applies each model’s coordinate decoder to intermediate layer representations. Middle layers produce representations furthest from the final output (highest RMSD), while early-layer representations are closer — the network maps through an internal space not aligned with coordinate space. Contact similarity to the final structure is near zero until the last 1–2 layers, concentrating structural specificity at the output. The radius of gyration panel shows that intermediate layers decode to structures with varying compactness; the trajectory from noise to structure is not monotonic through the layer stack. Triangle models show a more gradual RMSD gradient, consistent with the pair representation providing an additional channel for structural information.

4 Discussion

The geometric bias is essential late in generation, regardless of scale or architecture. The temporal asymmetry is our strongest finding because it is not predicted by B ’s construction. At $t=0.3$, the noisy structure $\mathbf{x}_t$ already has partial distance information that could in principle be useful for coarse-grained orientation. Yet the causal evidence shows the model gains nothing from early B that it cannot recover later. This suggests the model’s early-stage velocity field operates in a regime where geometric specificity is unnecessary — or even counterproductive — and that B ’s role is confined to late-stage refinement when the structure is close enough to final for fine-grained geometric guidance to be actionable.

Descriptive and causal measures tell different stories. The descriptive R_c profiles vary dramatically across models: R_c ranges from 0.5 to 35, and the peak layer varies from L0 to L9. Yet the causal ablation pattern is identical across all four models. This dissociation has methodological implications: conclusions about geometric bias importance drawn from magnitude-based metrics alone would suggest fundamentally different roles for B across scales, while the causal evidence reveals the same underlying dependence.

Notably, the *descriptive* peak- R_c layer does not predict which layers are *causally* critical. The 400M tri model concentrates $R_c = 29.4$ in Layer 2 (an early layer), but its causal profile matches the 200M tri model (deep layers critical, early layers dispensable). This underscores that signal magnitude is not a reliable proxy for causal importance.

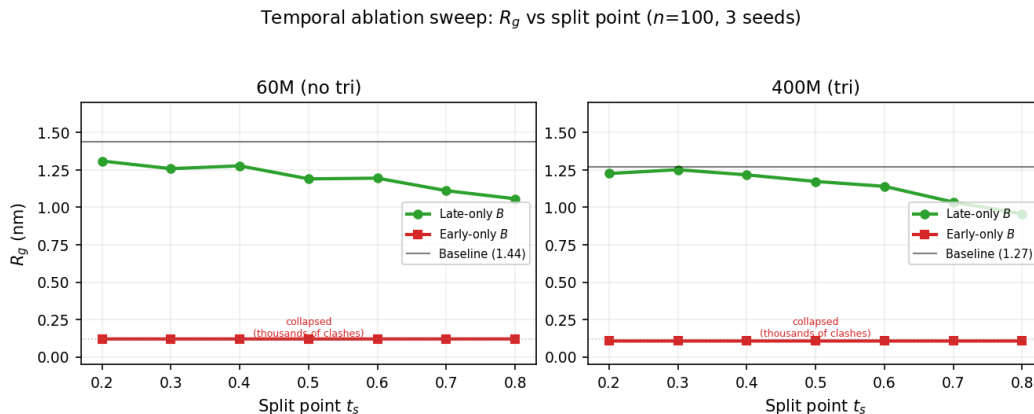


Figure 1: Fine-grained temporal sweep ($n=100$, 3 seeds). Early-only B (red) always collapses regardless of the split point. Late-only B (green) always produces valid structures with gradually decreasing quality. The asymmetry is one-sided, not a cliff at a specific timestep.

Triangle updates determine where B is critical, not when. Without triangle updates, Layer-0 ablation is most damaging and deep ablation is moderate. With triangle updates, this inverts. Triangle updates create a deep-layer information bottleneck through the pair representation: the iterative refinement of pair features in deep layers makes B ’s contribution there indispensable, while earlier layers become robust to missing geometric signal. Importantly, the temporal pattern is unaffected — triangle updates change the architecture’s spatial sensitivity to B without altering the generative process’s temporal dependence on it.

Limitations and future directions. Because all models were trained with B present, C may never have been pressured to encode structural information. The causal necessity we observe may partly reflect shortcut learning — the model adapted to B ’s availability during training — rather than a fundamental architectural limitation of content-only attention. Testing this would require training models without B , which is beyond the scope of this study. All functional analyses use retrospective ground truth (the model’s own output), making the B -vs- C contact precision comparison partly tautological at late timesteps. Extending to conditional generation, where ground truth contacts come from experimentally determined structures, would break this circularity and test whether the temporal and causal findings hold when external structural information is available.

5 Conclusion

The geometric pair bias in Proteina is dispensable during early generation but essential during late structural refinement — a finding that replicates across four model scales, three protein lengths, and is not a gradual transition but a one-sided asymmetry. The model depends on the specific geometric *information* in B , not its magnitude. Triangle updates alter *where* in the network B is critical without changing *when*. Descriptive signal magnitude is decoupled from causal necessity, and the descriptive peak layer does not predict the causally critical layers — multi-lens interpretability is necessary to distinguish these.

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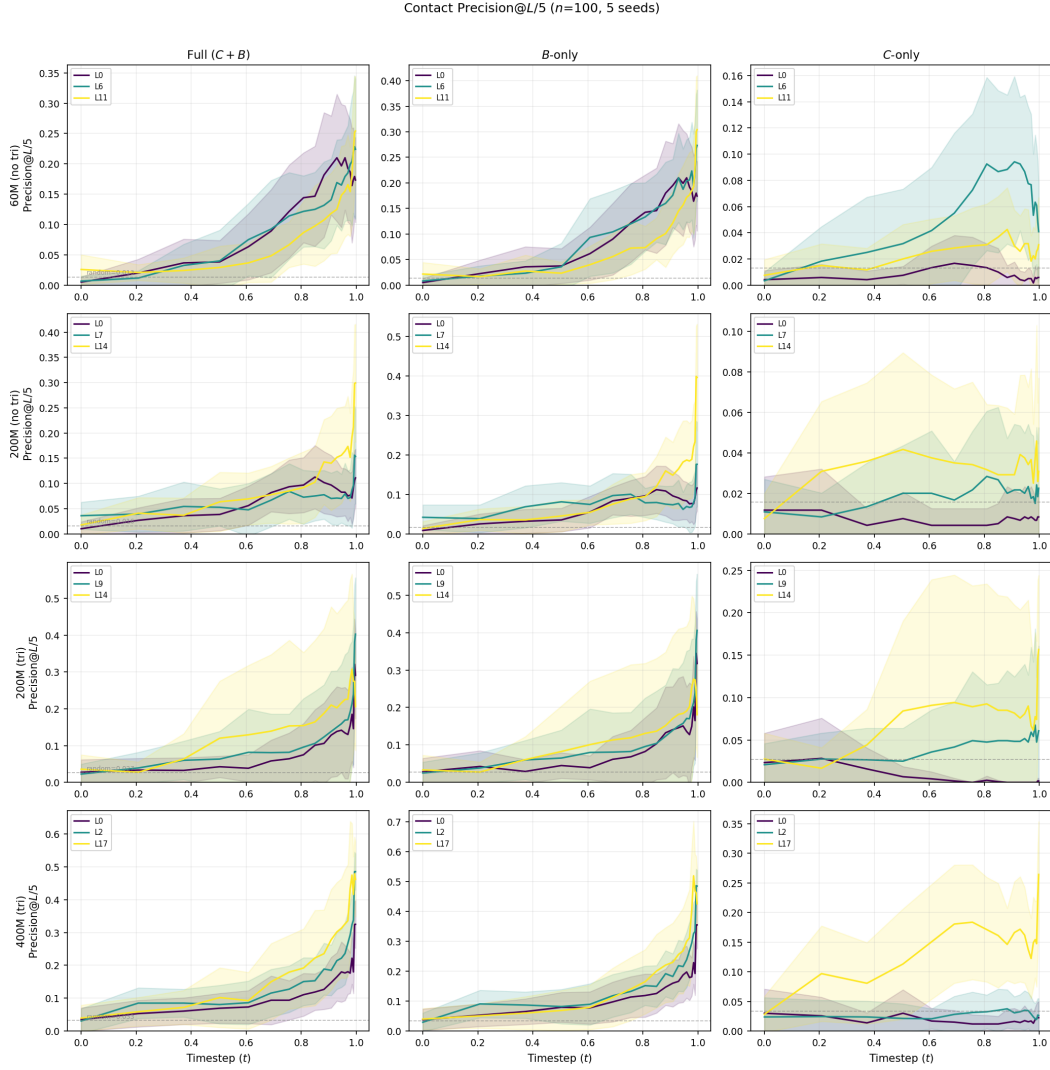


Figure 2: Contact Precision@L/5 ($n=100$, 5 seeds). Dashed line: random baseline. *B*-only outperforms *C*-only at all scales, but this is partly expected by construction (see text).

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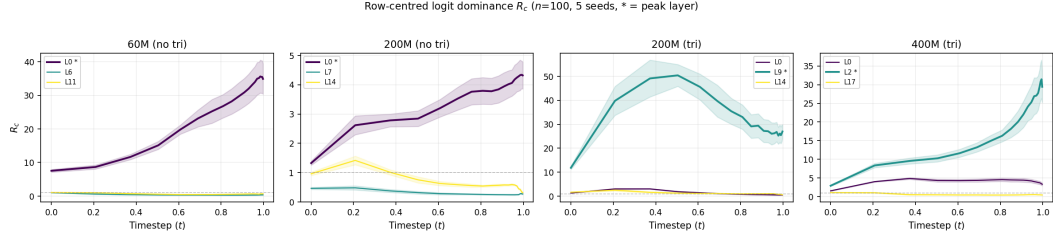


Figure 3: R_c for key layers ($n=100$, 5 seeds). Peak R_c layer marked with *. Dashed line: $R_c = 1$.

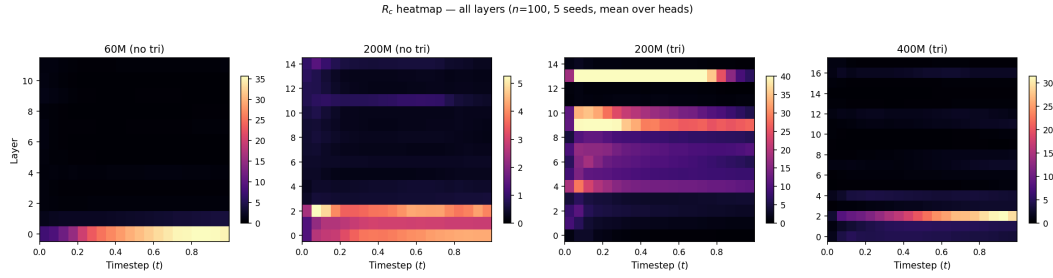


Figure 4: R_c across all layers and timesteps (mean over heads). Each model concentrates geometry dominance differently: L0 in 60M, L0/L2 in 200M no-tri, distributed across many layers in 200M tri, L2 in 400M tri.

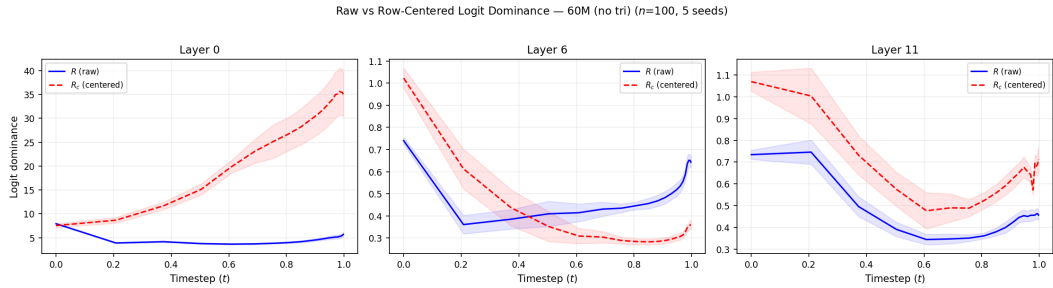


Figure 5: Raw R vs. row-centred R_c for the 60M model. R_c diverges from R at late timesteps: C develops softmax-invariant energy.

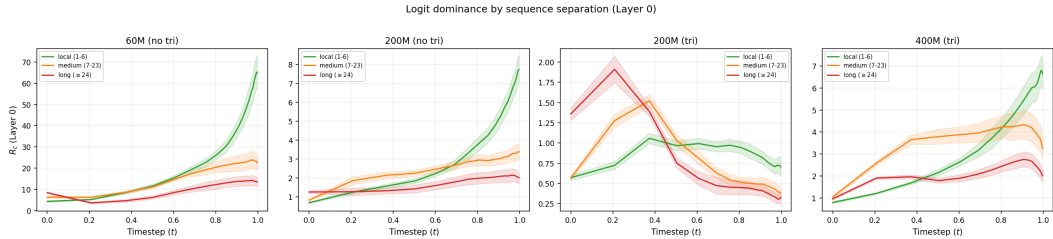


Figure 6: R_c by sequence separation in Layer 0 for all four models (5 seeds).

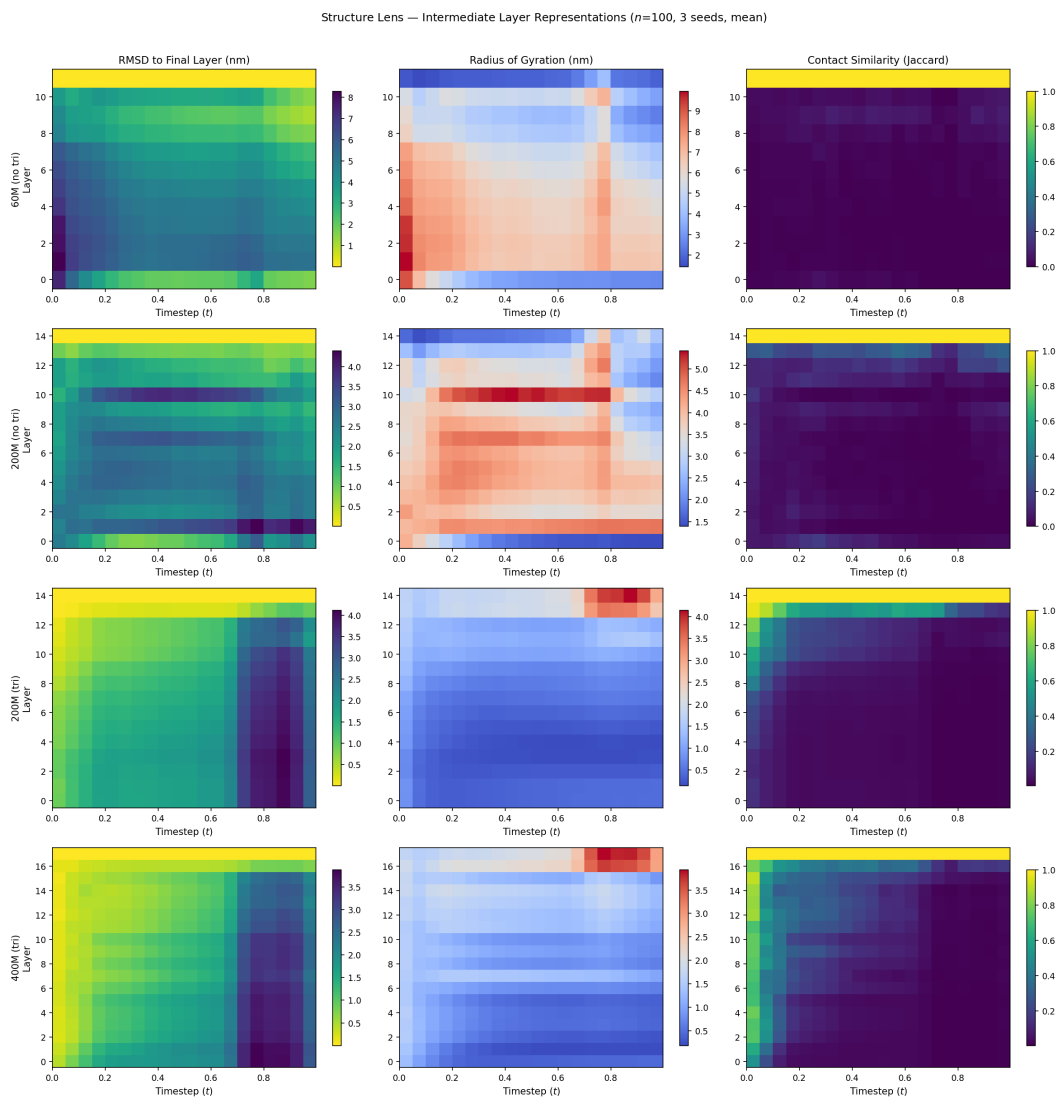


Figure 7: Structure lens (3 seeds, mean). **Left:** RMSD to final-layer prediction. **Centre:** Radius of gyration. **Right:** Contact similarity (Jaccard, $< 8 \text{ \AA}$, $|i-j| \geq 6$).